

Zerui Guo

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I build hardware-aware systems for efficient, predictable execution across accelerators, networks, and storage.

Education

University of Wisconsin–Madison	Expected May 2027
Ph.D. in Computer Science; Advisor: Ming Liu	Jun. 2023–Present
Beihang University	Jan. 2023
Master of Computer Science and Technology; Advisor: Yuebin Bai	
Beijing University of Posts and Telecommunications	Jun. 2020
Bachelor of Network Engineering; ranked 4/135	

Research Projects

MemChannel: Transport for Switched CXL Memory Pooling	Jun. 2023–Apr. 2026
• Built a CSFQ-inspired transport for max-min fair bandwidth sharing among competing flows in switched CXL memory pools, using a fabric-level fluid model and congestion feedback to adapt per-flow access rates. • Implemented host-side admission control with POSIX timers and CXL performance counters, coordinating rates across hosts and supporting unmodified applications through library injection. • Accelerated scientific and graph workloads by $2.0\text{--}3.5\times$ under heavy contention while sustaining over 96% memory-bandwidth utilization through host- and fabric-level performance isolation.	
LogNIC: Performance Modeling for SmartNICs	Feb. 2022–Apr. 2023
• Implemented a Python model of packet-centric execution graphs to predict latency/throughput and diagnose compute and data-movement bottlenecks; validated on LiquidIO-II, BlueField-2, and Stingray SmartNICs. • Built a SciPy-based optimizer for core allocation and compute placement under bandwidth and latency constraints. • Model-guided SmartNIC core allocation increased throughput by 36.4% and reduced average request latency by 22.8% versus equal partitioning, averaged across five microservice workloads.	
LEED: Persistent Key-Value Storage on SmartNICs	Sep. 2021–Feb. 2023
• Designed and implemented a replicated persistent key-value store on Stingray SmartNIC JBOFs using SPDK/NVMe, supporting node joins and fail-stop recovery through data migration and chain reconfiguration. • Implemented circular SSD logs, DRAM/flash hybrid indexing, and token-based I/O scheduling with backpressure. • Delivered $4.0\times$ the energy efficiency of server-based storage and $18.3\times$ that of embedded storage, measured in requests per joule and averaged across six workloads and two object sizes.	

Selected Publications

[1] Building A CSFQ-Inspired Transport for Switched CXL Memory Pooling	NSDI '26
<u>Zerui Guo</u> , Emily Shriver, Ming Liu.	
[2] Understanding and Profiling CXL.mem Using PathFinder	SIGCOMM '25
Xiao Li, <u>Zerui Guo</u> , Yuebin Bai, Mahesh Ketkar, Hugh Wilkinson, Ming Liu.	
[3] Building Massive MIMO Baseband Processing on a Single-Node Supercomputer	NSDI '25
Xincheng Xie, Wentao Hou, <u>Zerui Guo</u> , Ming Liu.	
[4] LogNIC: A High-Level Performance Model for SmartNICs	MICRO '23
<u>Zerui Guo</u> , Jiaxin Lin, Yuebin Bai, Daehyeok Kim, Michael Swift, Aditya Akella, Ming Liu.	
[5] LEED: A Low-Power, Fast Persistent Key-Value Store on SmartNIC JBOFs	SIGCOMM '23
<u>Zerui Guo</u> , Hua Zhang, Chenxingyu Zhao, Yuebin Bai, Michael Swift, Ming Liu.	

Systems Engineering Experience

EFabric — Distributed Storage Infrastructure

Feb. 2026–Present

University of Wisconsin–Madison

- Built and now maintain a 3 PB raw-capacity SeaweedFS/JuiceFS storage cluster for EFabric, an AI-for-science computing platform serving hundreds of users and running dozens of experiments daily.
- Built tooling to rebuild corrupted object mappings and recover files with checksum and payload validation.
- Diagnosed and reported three SeaweedFS correctness bugs in metadata initialization, volume balancing, and compaction; supplied root-cause analyses and a deterministic corruption reproducer, leading to upstream fixes.

Alibaba Cloud — Research Intern

May–Aug. 2024

Bellevue, Washington

- Designed and prototyped a BlueField-3-centric storage engine for ZNS NVMe SSDs, targeting host-CPU offload for cloud chunk storage through SPDK-based zoned-device access and asynchronous I/O.

Professional Service

Reviewer, IEEE Transactions on Computers

2026

Reviewer, ACM Transactions on Architecture and Code Optimization

2024

Technical Skills

Languages: C/C++, Python, Rust, Go, JavaScript

Systems: Linux systems programming, RDMA, UCX, CXL, SPDK, NVMe